CUSTOMER BEHAVIOUR ANALYSIS

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This project involved analyzing customer shopping behavior based on historical purchase patterns using a dataset of around 3,900 records. Data cleaning, transformation, and analysis were performed using Python (Pandas) and SQL Server. Power BI dashboards were created to visualize key insights such as customer segments, purchasing trends, and revenue contribution. The findings helped management understand customer behavior and supported strategic business decisions.

Problem Statement

A leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. The management team has noticed changes in purchasing patterns across demographics, product categories, and sales channels (online vs. offline). They are particularly interested in uncovering which factors, such as discounts, reviews, seasons, or payment preferences, drive consumer decisions and repeat purchases.

You are tasked with analyzing the company's consumer behavior dataset to answer the following overarching business question:

"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

Questions Asked by management:

1) What is the total revenue generated by male versus female customers?
2) Which customers used a discount but still spent more than the average purchase amount?
3) Which are the top five products with the highest average review rating?
4) How does the average purchase amount compare between standard and express shipping?
5) Do subscribed customers spend more than non-subscribed customers?
6) Which five products have the highest percentage of purchases with discounts applied?
7) How can customers be segmented into new, returning, and loyal based on the total number of previous purchases?
8) What are the top three most purchased products within each category?
9) Are customers who are repeat buyers (with more than five previous purchases) also likely to subscribe?

10) what is the revenue contribution of each age group?

Tools Used:

- Python Pandas for data cleaning
- SQL server for data manipulation
- Power BI for Interactive dashboard

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Process Followed:

1)On Pandas checked whether on Data is there any null values or not?

```
[14]: df.isnull().sum()
```

```
[14]: Customer ID                0
      Age                        0
      Gender                     0
      Item Purchased             0
      Category                   0
      Purchase Amount (USD)      0
      Location                   0
      Size                       0
      Color                      0
      Season                     0
      Review Rating             37
      Subscription Status        0
      Shipping Type              0
      Discount Applied           0
      Promo Code Used            0
      Previous Purchases         0
      Payment Method             0
      Frequency of Purchases     0
      dtype: int64
```

Found 37 Null values in Review Ratings

2) Checked statistical conditions of datas:

```
[16]: df.describe()
```

[16]:

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

3) Converted all records in data frame in Lower case to ensure smooth process

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```
df.columns = df.columns.str.strip().str.lower().str.replace(' ', '_')
```

```
df.columns = df.columns.str.replace(' ', '_')
```

4) Handled null values by filling it with median of Review ratings

```
df['review_rating'] = df.groupby('category')['review_rating']\
                        .transform(lambda x: x.fillna(x.median()))
```

5) Found 2 columns with same records, ensured it and delete one column

```
(df['discount_applied'] == df['promo_code_used']).all()
```

```
np.True_
```

```
df.drop('promo_code_used',axis=1)
```

6) According to age it is divided into equal 4 groups, 'young adult', 'adult', 'middle aged', 'senior' and added new column

```
labels=['young_adult','adult','middle_aged','senior']
df['age_group']=pd.qcut(df['age'],q=4,labels=labels)
```

```
df[['age','age_group']].head(10)
```

	age	age_group
0	55	middle_aged
1	19	young_adult
2	50	middle_aged
3	21	young_adult
4	45	middle_aged
5	46	middle_aged
6	63	senior
7	27	young_adult
8	26	young_adult
9	57	middle_aged

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7) Added new column frequency_of_purchases by distinguish into 'Fortnightly', 'Weekly', Bi-Weekly', 'Monthly', 'Every 3 months', 'Quarterly', 'Annually'

```
frequency_mapping={'Fortnightly':14,'Weekly':7,'Monthly':30,'Quarterly':90,'Bi-Weekly':14,'Annually':365, 'Every 3 Months':90}
df['purchase_frequency_days']=df['frequency_of_purchases'].map(frequency_mapping)
```

```
df[['purchase_frequency_days','frequency_of_purchases']].head(10)
```

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

8)Connected my jupyter notebook to sql server and exported data to Database name as 'Customer_Behaviour' and table name as 'customer'

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```
import pyodbc
pyodbc.drivers()
```

```
['SQL Server',
 'Microsoft Access Driver (*.mdb, *.accdb)',
 'Microsoft Excel Driver (*.xls, *.xlsx, *.xlsm, *.xlsb)',
 'Microsoft Access Text Driver (*.txt, *.csv)',
 'SQL Server Native Client RDA 11.0',
 'ODBC Driver 17 for SQL Server']
```

```
from sqlalchemy import create_engine •••
```

```
pd.read_sql("SELECT @@VERSION AS version", engine)
```

	version
0	Microsoft SQL Server 2022 (RTM) - 16.0.1000.6 ...

```
df.to_sql(
    name='customer',
    con=engine,
    schema='dbo',
    if_exists='replace',
    index=False
)
```

As I had finished Data cleaning now here are the answers for the questions by management

1)**What is the total revenue generated by male versus female customers?**

```
select gender,sum([purchase_amount_(usd)]) as revenue_usd
from customer
group by gender
```

Results | Messages

	gender	revenue_usd
1	Male	157890
2	Female	75191

2) **Which customers used a discount but still spent more than the average purchase amount**?

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```
select customer_id,[purchase_amount_(usd)]
from customer
where discount_applied='yes' and [purchase_amount_(usd)]>=(select avg([purchase_amount_(usd)]) from customer)
```

	customer_id	purchase_amount_(usd)
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88

Found total 863 customers used discount but spend more than average purchase price

3) Which are the top five products with the highest average review rating?

```
select top 5
item_purchased,round(avg(review_rating),2) as top_5_products_average_rating
from customer
group by item_purchased
order by top_5_products_average_rating desc
```

	item_purchased	top_5_products_average_rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.8
5	Handbag	3.78

4) How does the average purchase amount compare between standard and express shipping?

```
select shipping_type,
avg([purchase_amount_(usd)])as average_amount from customer
where shipping_type in('standard','express')
group by shipping_type
```

	shipping_type	average_amount
1	Standard	58
2	Express	60

5) Do subscribed customers spend more than non-subscribed customers?

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```sql
select subscription_status,count(customer_id) as number_of_customer,
sum([purchase_amount_(usd)]) as total_revenue,
avg([purchase_amount_(usd)]) as average_revenue
 from customer
group by subscription_status;
```

	subscription_status	number_of_customer	total_revenue	average_revenue
1	Yes	1053	62645	59
2	No	2847	170436	59

6) Which five products have the highest percentage of purchases with discounts applied?

```sql
select top 5
item_purchased,
ROUND(100* sum(case when discount_applied ='yes' then 1 else 0 end)/count(*),2) as discount_rate
from customer
group by item_purchased
order by discount_rate desc
```

	item_purchased	discount_rate
1	Hat	50
2	Coat	49
3	Sneakers	49
4	Sweater	48
5	Pants	47

7) 7) How can customers be segmented into new, returning, and loyal based on the total number of previous purchases?

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```sql
with customer_type as(
select customer_id,previous_purchases,
case
when previous_purchases=1 then 'new'
when previous_purchases between 2 and 10 then 'returning'
else 'loyal'
end as customer_segment
from customer)
select customer_segment, count(*) as number_of_customer
from customer_type
group by customer_segment
```

	customer_segment	number_of_customer
1	returning	701
2	loyal	3116
3	new	83

8) What are the top three most purchased products within each category?

```sql
with item_count as (
select category,item_purchased ,count(customer_id) as total_orders,
row_number() over (partition by category order by count(customer_id)desc)as item_rank
from customer
group by category,item_purchased)

select item_rank,category,item_purchased,total_orders from item_count
where item_rank<=3;
```

	item_rank	category	item_purchased	total_orders
1	1	Accessories	Jewelry	171
2	2	Accessories	Belt	161
3	3	Accessories	Sunglasses	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9) Are customers who are repeat buyers (with more than five previous purchases) also likely to subscribe?

```
select subscription_status,count(customer_id) as repeat_buyers
from customer
where previous_purchases>5
group by subscription_status
```

	subscription_status	repeat_buyers
1	Yes	958
2	No	2518

10) what is the revenue contribution of each age group?

```
select age_group,
sum([purchase_amount_(usd)]) as total_revenue
from customer
group by age_group
order by total_revenue desc
```

	age_group	total_revenue
1	young_adult	62143
2	middle_aged	59197
3	adult	55978
4	senior	55763

Insights And Recommendation:

After analyzing the data here are the insights I had found that help to boost the sales:

- Discount attract traffics, need to be careful at margin but discount can be used as tools to boost average purchase amount
- Subscribing status is not making any change in shopping habits, need to review it
- Adult and middle aged are contributing high in revenue, can run marketing on this age group
- Faster the delivery more revenue, need to work on it to make it fast

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